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GitX1

A Hierarchical Defense-in-Depth Architecture for Mitigating AI-Generated Pull Request Pollution in Open-Source Ecosystems

Keywords: *LLM pollution • AI-generated code • pull request filtering • WebAssembly • AST fingerprinting • open-source ecosystems • verification tax • throughput asymmetry*

Abstract

The advent of Large Language Models (LLMs) specialized in software development has fundamentally altered the open-source software (OSS) contribution landscape. While these tools lower the barrier to entry, they have inadvertently catalyzed a surge in low-quality, AI-generated pull requests (PRs) often colloquially referred to as "slop." In this paper, we formalize this threat through the lenses of throughput asymmetry and the verification tax, illustrating how the speed of AI generation drastically outpaces human review capacity while imposing a heavy cognitive load to identify plausible but fundamentally flawed code. To address this, we introduce GitX1, a hierarchical, defense-in-depth architecture implemented as a client-side Chrome extension and powered by WebAssembly (Wasm). GitX1 applies a multi-layered filtering strategy: (1) syntactic and linguistic heuristic engines, (2) contributor reputation scoring, and (3) deep structural AST and pattern analysis via Wasm modules. By categorizing and flagging AI-generated PR pollution natively within the GitHub interface, GitX1 restores maintainer agency, mitigates the verification tax, and helps preserve the integrity of open-source ecosystems.

1. Introduction

Open-source software (OSS) ecosystems rely on a delicate socio-technical balance: the continuous influx of community contributions is matched by the rigorous, often volunteer-run, review processes of project maintainers. Historically, this balance was governed by the natural limits of human developer productivity. However, the proliferation of increasingly capable Large Language Models (LLMs) and autonomous coding agents has disrupted this equilibrium. Today, contributors can generate vast amounts of code with minimal effort, resulting in a dramatic rise in AI-generated pull requests (PRs).

While many of these tools aim to enhance productivity, they have concurrently enabled a new class of contribution pollution, saturating repositories with structurally plausible but functionally or semantically flawed submissions. This emerging crisis can be understood through two core concepts: throughput asymmetry and the verification tax. The real-world consequences are already observable, evidenced by the temporary shutdown of the Jazzband Python collective due to overwhelming AI-generated "slop" [1], and the cancellation of the cURL project's bug bounty program following an influx of fabricated vulnerability reports [2].

Core Problem: Throughput Asymmetry [3]

An LLM-backed agent can submit dozens of pull requests across multiple repositories in the time it takes a human maintainer to thoroughly review a single file. This asymmetry weaponizes scale, turning high-volume submission capabilities into a denial-of-service vector against human attention.

Consequently, maintainers are subjected to an unprecedented Verification Tax. Unlike traditional spam, AI-generated code pollution is deceptively coherent. It often mimics repository style guidelines and produces plausible documentation. Assessing whether these submissions contain subtle logical flaws, hallucinatory API calls, or security vulnerabilities requires deep cognitive engagement. The verification tax is the disproportionate cognitive load imposed on maintainers to review this "plausible slop," rapidly accelerating maintainer burnout and degrading overall project health.

To combat this, we present GitX1, a novel defense-in-depth architectural intervention designed to abate AI-generated PR pollution natively within the maintainer's workflow. GitX1 employs a multi-tiered defense strategy:

- Layer 1 Triage & Heuristic Engines: Lightweight TypeScript engines analyzing burstiness, linguistic artifacts, and structural anomalies.
- Reputation & Behavioral Scoring: Background analysis of contributor history to identify hit-and-run agentic behavior.
- Deep Structural Analysis (Wasm): A Rust-based WebAssembly module performing deep AST validation and diff-parsing without server-side compute.

By fusing these hierarchical layers, GitX1 provides real-time, explainable scoring and triage directly at the PR review surface.

2. Methodology and System Architecture

To counter throughput asymmetry without compounding the verification tax, GitX1 is built as a cascading, client-side defense pipeline. Operating natively within the browser as a Chrome extension, the architecture avoids the latency, cost, and privacy concerns of server-side evaluation. Our methodology is defined by a four-layer sequential vetting process, where each layer increases in computational depth and specificity.

Table 1. *GitX1 Four-Layer Architecture — Technology and Detection Signal Summary*

Layer	Name	Technology	Detection Signal	Complexity
Layer 1	**Instant Triage**	TypeScript (Content Scripts)	Metadata burstiness, commit clustering, PR velocity	O(1) — always-on
Layer 2	**Reputation Scoring**	GitHub API + Service Workers	Account age, PR acceptance rate, fork timing	O(n) API calls
Layer 3	**AST Fingerprinting**	Rust → WebAssembly	Tree-sitter AST, diff parsing, semantic anomalies	O(k) per diff hunk
Layer 4	**AI Text Detection**	7 Local NLP Engines	Burstiness, vocabulary, formulaic patterns, sentiment	O(m) per description

2.1 System Execution Flow

Figure 1 illustrates the zero-trust evaluation pipeline and the client-side execution sequence from the moment a Pull Request is opened. Each decision gate corresponds to one of the four defense layers; a rejection at any gate terminates further processing and immediately renders the verdict in the GitHub UI.

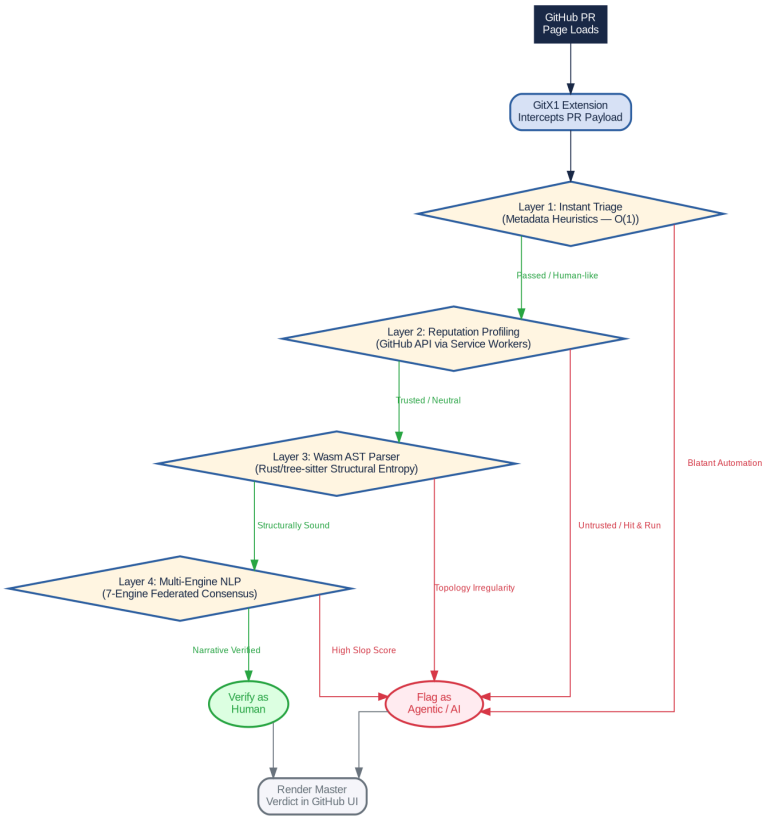

Figure 1. *GitX1's cascaded four-layer architecture pipeline executing natively at the browser edge. Red paths terminate with an agentic flag; green paths advance to deeper scrutiny.*

2.2 Layer 1: Instant Triage (Language-Agnostic Metadata)

The first line of defense is a high-speed, language-agnostic triage mechanism designed to filter blatant automation artifacts. Operating at the interaction layer via content scripts, this module evaluates PR metadata such as submission velocity, description length relative to code changeset, and commit timestamp clustering. Because AI agents often generate massive PRs with minimal time delta between cloning, branch creation, and PR submission, this layer rapidly identifies anomalous burstiness. The O(1) complexity ensures imperceptible latency on the review interface across all 29 heuristic checks.

2.3 Layer 2: Contributor Reputation Scoring

Submissions that bypass initial triage are subjected to contributor reputation scoring. Leveraging the GitHub API through background service workers, this layer computes a heuristic trust score based on historical and behavioral metrics:

- Account Age & Activity Vectors: Flagging newly created or dormant accounts exhibiting sudden spikes in contribution volumes.
- PR Acceptance Rate: Evaluating the historical ratio of merged versus closed/rejected PRs as a signal of authentic human contribution.
- Fork Timing Analysis: Measuring latency between forking an upstream repository and submitting a patch to identify hit-and-run agentic behaviors.

This layer provides a contextual, dynamic baseline for the submitter, weighting the necessity for deeper subsequent analysis through a numeric reputation score ranging from −25 to +15.

2.4 Layer 3: Structural AST Fingerprinting (WebAssembly)

To detect the structural anomalies characteristic of LLM hallucinations such as cyclical logic, importing non-existent external libraries, or syntactically correct but semantically nonsensical patterns, GitX1 employs advanced Abstract Syntax Tree (AST) fingerprinting. We introduce a novel approach by compiling Rust-based tree-sitter logic into WebAssembly (Wasm). Executed entirely within the browser's execution sandbox, the gitx1-wasm module parses incoming diffs and performs zero-latency structural analysis, ensuring even obfuscated AI-generated code is evaluated topologically rather than purely textually.

2.5 Layer 4: Multi-Engine AI Text Detection

The final layer scrutinizes the natural language components of the PR, including titles, descriptions, and code comments. Architected as a multi-engine consensus system, this layer runs seven lightweight, specialized heuristic models natively:

- Burstiness & Vocabulary Engines: Measuring n-gram diversity, linguistic distribution, and the predictability inherent to transformer-based generation.
- Formulaic & Structural Engines: Detecting repetitive LLM boilerplate such as standard disclaimer phrases, overly rigid bulleted lists, and formulaic apologies.
- Linguistic & Sentiment Engines: Analyzing tone uniformity and unnatural syntactic cadences characteristic of autoregressive generation.

The system's baseline metrics are calibrated against the HMCorp dataset [4], which contains 550,000 pairs of human-written and ChatGPT-generated code, alongside the AIDev dataset [5], which contains 33,707 agent-authored pull requests. By aggregating probabilistic outputs locally, GitX1 provides maintainers with a unified, explainable "slop score" ensuring strict data privacy.

3. Empirical Evaluation and Case Study

To validate the efficacy of GitX1's multi-layered defense architecture, we analyzed a real-world test case encountered in the wild. This instance illustrates the subtle nature of AI-generated repository pollution and the limitations of single-layer evaluation heuristics.

3.1 Case Study: Pull Request #662

We evaluated Pull Request #662 submitted to the repository KathiraveluLab/Beehive authored by @Pragya-rathal. The submission was processed through the GitX1 extension pipeline with results detailed in Table 2.

Table 2. *GitX1 Layer-by-Layer Evaluation Results — PR #662 (KathiraveluLab/Beehive)*

Layer	Score	Verdict	Key Finding
Layer 1 — Instant Triage	**100 / 100**	**CLEAN**	29/29 checks passed
Layer 2 — Reputation Scoring	**+10**	**ALLOW / Trusted**	Account age, PR rate, activity positive
Layer 3 — AST Fingerprinting	**100 / 100**	**CLEAN**	Zero structural anomalies detected
Layer 4 — AI Text Detection	**33.8 / 100**	**SUSPICIOUS**	Burstiness 94%, Readability 91%
MASTER VERDICT	**25.9**	**SUSPICIOUS**	Escalate — AI-authored description

The results reveal a critically important pattern. Layers 1 through 3 all returned clean scores: Layer 1 cleared all 29 metadata heuristics (100/100), Layer 2 assigned a positive reputation score (+10) based on account age, acceptance rate, and activity history, and Layer 3's Wasm AST parser found zero structural anomalies in the modified file.

However, Layer 4's seven-engine NLP consensus identified unmistakable LLM-generation signatures within the PR description. The burstiness engine flagged a 94% anomaly score, the readability engine flagged 91%, and sentiment uniformity reached 60%, yielding a combined Layer 4 score of 33.8/100 — SUSPICIOUS.

3.2 Master Verdict and Discussion

Aggregating scores across the complete pipeline, GitX1 delivered a final Master Verdict of 25.9 (Suspicious). This case study is deeply illustrative of why a multi-layered defense-in-depth architecture is strictly necessary in the modern OSS ecosystem.

Key Insight: Why Single-Layer Evaluation Fails

PR #662 passed Layers 1, 2, and 3 with perfect scores. The code was syntactically valid, the contributor appeared reputable, and the AST contained no structural anomalies. Only the multi-engine NLP consensus in Layer 4 — detecting burstiness (94%) and abnormal readability uniformity (91%) in the PR description — successfully identified the artificial origin of the contribution.

By capturing this discrepancy, GitX1 successfully mitigates the verification tax, automatically highlighting that despite the plausible code and trusted account, the narrative wrapper was artificially constructed, warning maintainers to approach the review with elevated caution.

4. Discussion: Privacy and Scalability

A crucial architectural decision in the design of GitX1 is its strict adherence to a client-side execution model. Implementing the entire four-layer defense pipeline entirely within the browser extension provides profound privacy and operational benefits.

Traditional automated analysis tools often require repository integration or demand that code diffs and PR descriptions be transmitted to third-party servers. In enterprise contexts, offloading proprietary code, unreleased security patches, or closely guarded algorithms to external APIs introduces unacceptable data exfiltration risks. By executing all structural fingerprinting and heuristic analysis locally, GitX1 guarantees a zero-trust privacy posture: proprietary code and contributor data never leave the reviewer's machine.

4.1 Graceful Degradation and Failure Handling

To satisfy enterprise reliability SLAs, GitX1 implements a robust circuit-breaker architecture enabling graceful degradation. Should the Layer 3 WebAssembly AST parser encounter memory constraints on abnormally monolithic PR diffs, or if the Layer 4 NLP ensemble experiences execution timeouts, the pipeline immediately traps the fault. Figure 2 illustrates the complete degradation sequence.

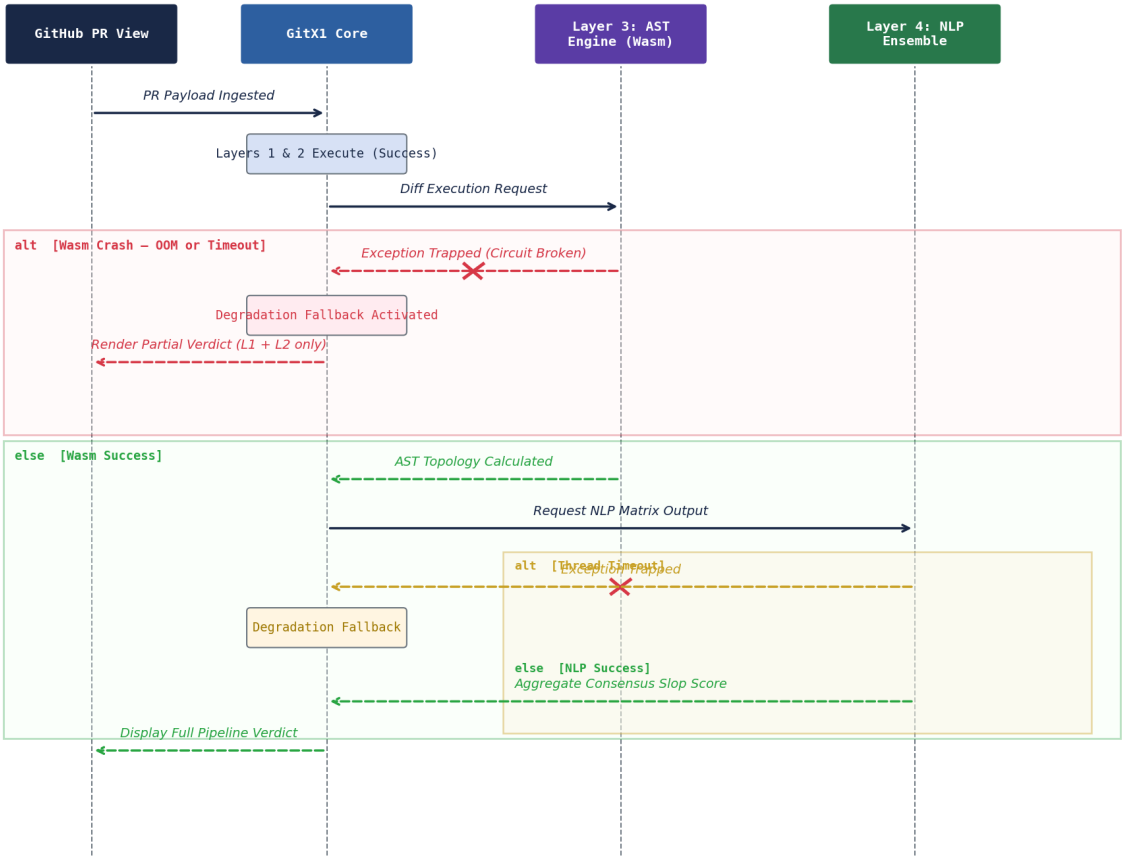

Figure 2. *GitX1 graceful degradation sequence diagram. On Wasm or NLP failure, the circuit-breaker traps the exception and falls back to Layers 1 and 2, ensuring tool failures never obstruct the review workflow.*

Rather than halting the code review interface, GitX1 seamlessly degrades to the O(1) metadata heuristic validations of Layer 1 and the deterministic reputation profiling of Layer 2. This strict fail-open methodology ensures that transient local outages or edge-case parsing panics never permanently obstruct legitimate human workflow.

4.2 Limitations and Future Work

GitX1's current heuristic baselines reflect the behavioral signatures of LLMs prevalent at the time the HMCorp and AIDev datasets were collected. As generative models grow more sophisticated in mimicking human writing patterns, continuous retraining and baseline updates will be necessary. Future work will investigate federated learning approaches that allow GitX1 instances to share anonymized detection signal updates without centralizing sensitive PR content.

Additionally, the current implementation operates as a read-only advisory layer. Planned extensions include integration with repository-level configuration to enable automated labeling or draft-conversion of flagged PRs, subject to explicit maintainer opt-in, to further reduce the verification tax without removing human agency from the review process.

5. Conclusion

The proliferation of LLMs has democratized software development, but it has simultaneously inverted the economic model of code review. The throughput asymmetry between autonomous submission agents and human maintainers has materialized as an unsustainable verification tax — a cognitive burden threatening the viability of open-source stewardship.

GitX1 addresses this existential threat by shifting the defense surface directly to the maintainer's browser. Through its hierarchical, defense-in-depth architecture combining instant metadata triage, reputation scoring, structural WebAssembly AST fingerprinting, and local multi-engine AI text consensus, GitX1 intercepts and flags highly plausible but artificially generated PR pollution that traditional checks miss. By functioning entirely client-side, the system respects the strict privacy and security mandates of modern software development.

The case study of PR #662 demonstrates concretely how each individual layer, while necessary, is insufficient in isolation: only the convergent, multi-layer pipeline correctly identifies the AI-generated origin of a contribution whose code is valid, whose author appears reputable, and whose structure is clean.

Summary

GitX1 restores the socio-technical balance of open-source ecosystems, actively protecting the fragile cognitive bandwidth of the stewards who maintain the world's digital infrastructure. By making AI-generated PR pollution visible at the point of review, it transforms an invisible, asymmetric threat into a tractable, manageable signal.

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